

Power BI – Athena Integration & Workspace Setup Summary

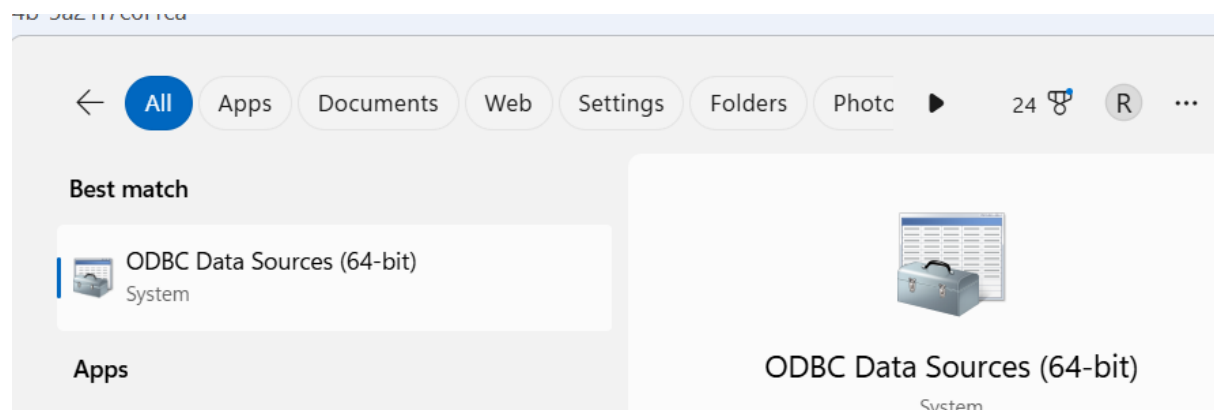
Objective

Connect Power BI with AWS Athena data, load datasets, and create a shared workspace so the team can collaboratively build and modify dashboards.

Work Completed (Step-by-Step Summary)

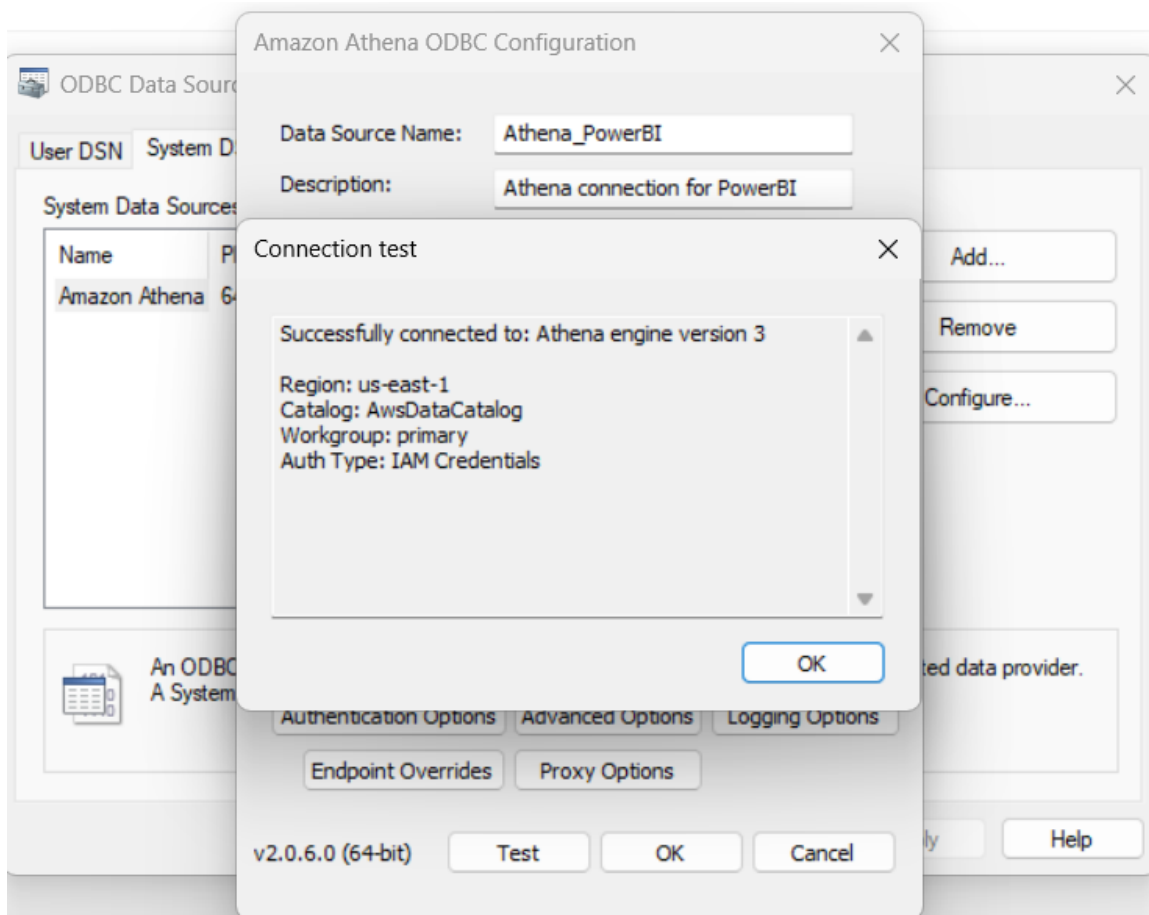
1. AWS Athena Connection Setup

- Configured Athena access using ODBC connection.
- Ensured team members received necessary IAM policies and credentials.
- Verified Athena catalogs and tables were accessible.
- Loaded required datasets into Power BI Desktop.



2. Dataset Loading in Power BI

- Connected Power BI Desktop to Athena via ODBC.
- Loaded analytics tables required for dashboards.
- Verified tables and columns loaded correctly.
- Previewed and validated data before loading.



3. Data Transformation & Validation

- Opened Power Query Editor.
- Loaded transformed data successfully into Power BI model.

Untitled - Power BI Desktop

File Home Help Table tools

Name final_master_produ...

Structure Manage relationships New measure Quick New table Mark as date table Calculations

parent_asin	main_category	category	brand	manufacturer	date_first_available	store	price	product_avg_rating	rating_number	total_reviews	review_avg_rating	rating
B01DSJXSDA	AMAZON FASHION	Unbranded			August 17, 2017			5	1	1	5	5
B01N661YB	AMAZON FASHION	Unbranded			December 12, 2016			5	1	1	5	5
B017XZBFUJ	AMAZON FASHION	Unbranded			November 9, 2015			5	1	1	5	5
B012DA3CQG	AMAZON FASHION	Unbranded			November 11, 2015			5	1	1	5	5
B01DX4F2K4	AMAZON FASHION	Unbranded			April 15, 2016			5	1	1	5	5
B01FVOSTS	AMAZON FASHION	Unbranded			December 28, 2016			5	1	1	5	5
B078VBM517	AMAZON FASHION	Unbranded			February 24, 2018			5	1	1	5	5
B01AJASSQJ	AMAZON FASHION	Unbranded			March 15, 2017			5	1	1	5	5
B07H3JN6JN	AMAZON FASHION	Unbranded			December 24, 2018			5	1	1	5	5
B07PH4KFMG	AMAZON FASHION	Unbranded			March 15, 2019			5	1	1	5	5
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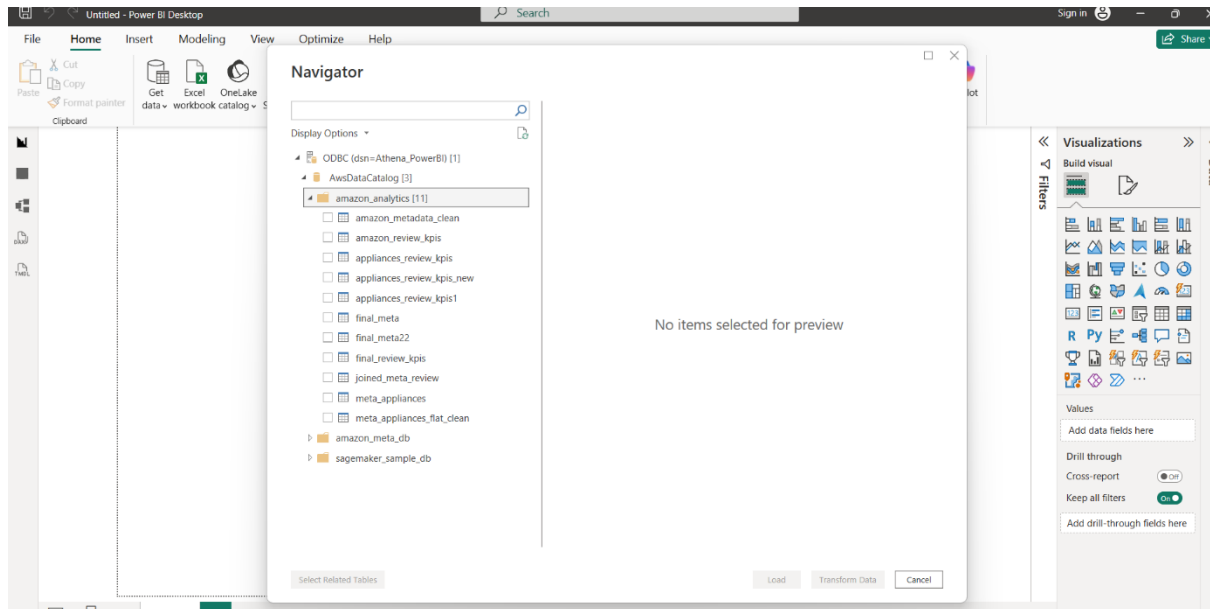
Data

Search

final_master_product_kpis

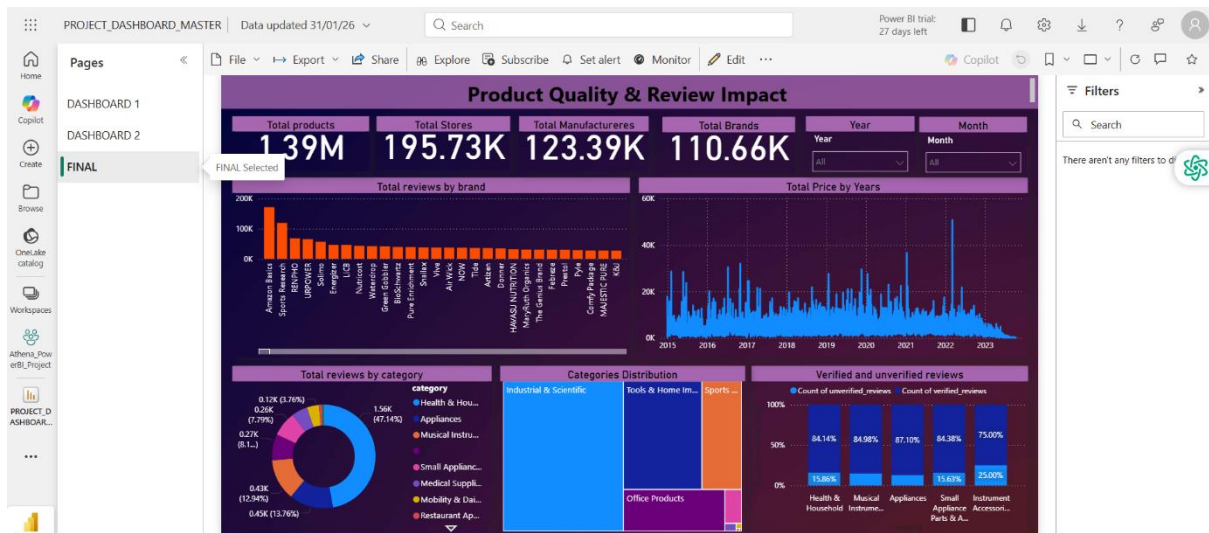
4. Dashboard File Integration

- Opened already created dashboard file shared by teammates.
- Confirmed visuals and tables were available.
- Linked dashboards to Athena dataset.



5. Dataset Finalization

- Verified all visuals functioned correctly with loaded dataset.
- Decided to keep date format unchanged to avoid rework.
- Ensured dataset was usable for dashboard development.



6. Power BI Account & Access Setup

- Logged into Power BI Service.
- Activated trial Power BI license.
- Resolved sign-in issues with organizational account.

7. Workspace Creation

- Created shared workspace:
Athena_PowerBI_Project

- Configured workspace for collaborative development.
 - Set license mode to Pro.
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8. Dashboard Publishing

- Published master dashboard from Power BI Desktop.
 - Verified report and semantic model appeared in workspace.
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9. Team Collaboration Enablement

- Added team members to workspace.
 - Assigned **Contributor role** so members can:
 - Edit dashboards
 - Create reports
 - Modify visuals
 - Publish updates
-

10. Workspace Ready for Team Usage

- Dashboard and dataset visible to team.
 - Collaborative development environment completed.
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Challenges Faced During Setup

1. Athena Access & Policy Issues

- Missing IAM permissions initially.
- Policy identification confusion.

2. Dataset Loading Confusion

- Multiple tables & catalogs caused selection confusion.

3. Date Format Issue

- Date column loaded as text.
- Required discussion whether to fix or continue.

4. Power BI Sign-In Problems

- Organizational account requirement unclear.
- Personal email not accepted initially.

5. Workspace vs File Sharing Confusion

- Difference between sharing PBIX and using workspace unclear initially.

6. Ownership & Publishing Flow Confusion

- Who should publish master file was unclear.
- Needed clear owner decision.

7. App Inclusion Confusion

- Unsure whether to include report in app.

Final Outcome

- ✓ Athena connected
- ✓ Data loaded into Power BI
- ✓ Dashboard integrated
- ✓ Workspace created
- ✓ Dashboard published
- ✓ Team added for collaboration
- ✓ Ready for joint development

Current Status

Project ready for collaborative dashboard development.

Project Summary: Power BI Dashboard Development using AWS Athena

In this project, our objective was to build a collaborative Power BI reporting solution using data stored in AWS Athena and enable multiple team members to work on the same dashboard efficiently.

I was responsible for setting up the end-to-end environment and enabling team collaboration.

Step 1 – Data Source Integration

I configured connectivity between Power BI Desktop and AWS Athena using an ODBC connection. Necessary IAM permissions were ensured so queries could be executed and datasets retrieved successfully.

Step 2 – Data Loading and Validation

The required tables were loaded into Power BI. I verified schemas, column mappings, and ensured data was correctly ingested into the Power BI model.

Step 3 – Dashboard Integration

We already had an initial dashboard layout. I integrated this dashboard with the Athena dataset, ensuring visuals were functional and aligned with the dataset structure.

Step 4 – Power BI Service & Workspace Setup

I created a shared Power BI workspace and published the master report along with its semantic model (dataset). This allowed centralized access to reports and datasets.

Step 5 – Collaboration Enablement

Team members were added as Contributors, enabling them to collaboratively:

- Edit dashboards
- Add new pages
- Modify visuals
- Work on the same dataset

This ensured controlled collaboration while maintaining a single source of truth.

Step 6 – Workflow Governance

To avoid conflicts, we established a rule where only the owner publishes updates, while contributors modify visuals within the shared workspace.

Challenges Faced

During implementation, we encountered a few challenges:

- Initial IAM permission and access configuration issues for Athena.
- Data type mismatches during dataset loading.
- Understanding workspace-based collaboration versus sharing PBIX files.
- Power BI service sign-in and licensing setup challenges.
- Aligning dashboard visuals with Athena datasets without breaking reports.

These were resolved through iterative configuration and validation.

Final Outcome

The result was a fully functional collaborative Power BI environment where multiple team members could work simultaneously on dashboards backed by Athena data, ensuring centralized control, consistent datasets, and efficient teamwork.

One-line Interview Summary

“I set up the end-to-end Power BI–Athena integration and implemented a shared workspace model so multiple team members could collaboratively build dashboards using a centralized dataset.”

If you want next, I can also prepare:

- ✓ 30-second HR answer
- ✓ Technical interviewer deep-dive version
- ✓ Resume project description
- ✓ STAR-method behavioral version
- ✓ Interview questions they may ask from this project

Just tell me what you want next.

can 2 member work at the same time

Short answer: Yes — but not in real-time like Google Docs.

Let me explain clearly so you know how teamwork actually works in Power BI.

✓ Can two members work at the same time?

Yes, two or more members can work at the same time, but with some rules.

Power BI works on a **save-based collaboration**, not real-time collaboration.

□ How collaboration actually works

Two teammates can:

- Open the same report
- Edit visuals
- Add pages
- Modify charts
- Save changes

But:

- Whoever **saves last overwrites previous changes** if editing the same page.
-

✓ Safe way for team to work simultaneously

Best practice

Divide work by pages:

Member	Page Responsibility
Member A	Dashboard Page 1
Member B	Dashboard Page 2
You	Final Page / Model

This avoids conflicts.

⚠ What happens if both edit the same page?

Example:

1. Person A edits Page 1
2. Person B edits Page 1 simultaneously
3. Person A saves
4. Person B saves later

👉 Person B's version overwrites Person A's changes.

✓ Best team workflow

- ✓ Different pages per member
 - ✓ Inform team before editing shared visuals
 - ✓ Owner handles dataset changes
-

✓ What is ODBC?

ODBC (Open Database Connectivity) is a **standard interface** that allows applications (like Power BI, Excel, Python, etc.) to connect to different databases **using a common method**, without needing database-specific code.

In simple terms:

ODBC acts as a translator between an application and a database.

So Power BI can talk to Athena, SQL Server, MySQL, PostgreSQL, etc., through the same mechanism.

✓ Why ODBC is needed

Different databases speak different protocols. Instead of Power BI learning every database language:

- ODBC provides a **universal communication layer**.
- Applications use ODBC.
- ODBC talks to the database using proper drivers.

This avoids writing custom connectors for each database.

✓ Main Components of ODBC

1. Application

Example:

- Power BI
- Excel
- Python script
- Reporting tools

This is where queries originate.

2. ODBC Driver Manager

This sits between application and drivers.

Responsibilities:

- Loads the correct database driver
- Manages connections
- Routes requests

In Windows, this is part of the OS.

3. ODBC Driver

A driver is database-specific.

Examples:

- Athena ODBC driver
- SQL Server ODBC driver
- MySQL ODBC driver

Driver translates ODBC calls into database-specific language.

4. Database / Data Source

Final destination where data exists:

- AWS Athena
 - SQL Server
 - MySQL
 - PostgreSQL
 - Oracle
-

How ODBC Works (Step-by-Step Flow)

When Power BI connects to Athena:

Step 1

User selects ODBC connection in Power BI.

Step 2

Power BI sends query via ODBC API.

Step 3

Driver Manager loads Athena ODBC driver.

Step 4

Athena driver converts query into Athena-compatible format.

Step 5

Query runs in Athena.

Step 6

Results return through driver → ODBC → Power BI.

Step 7

Power BI displays data.

Flow:

Power BI



ODBC Manager



Athena ODBC Driver



AWS Athena



Results back to Power BI

✓ What is DSN (Data Source Name)?

DSN stores connection configuration.

It contains:

- Database endpoint
- Credentials
- Driver info
- Port
- Schema/catalog

So instead of configuring connection every time, Power BI just selects DSN.

Example:

Athena_PowerBI

✓ Why ODBC was used in your project

Athena doesn't have a native direct connector in all Power BI environments.

So:

- Athena ODBC driver installed
- DSN configured
- Power BI connected via ODBC

This allowed Athena queries from Power BI.

✓ **Advantages of ODBC**

- Works with many databases
 - Standard interface
 - Easy integration
 - Supported by most tools
 - No need for custom connectors
-

⚠ **Limitations**

- Sometimes slower than native connectors
 - Driver configuration required
 - Errors harder to debug
 - Driver compatibility issues
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✓ **Interview-Ready Answer (Short Version)**

If interviewer asks:

What is ODBC?

Answer:

ODBC is a standard database connectivity interface that allows applications like Power BI to connect to various databases using drivers. In our project, we used an Athena ODBC driver so Power BI could query AWS Athena data through a DSN configuration.

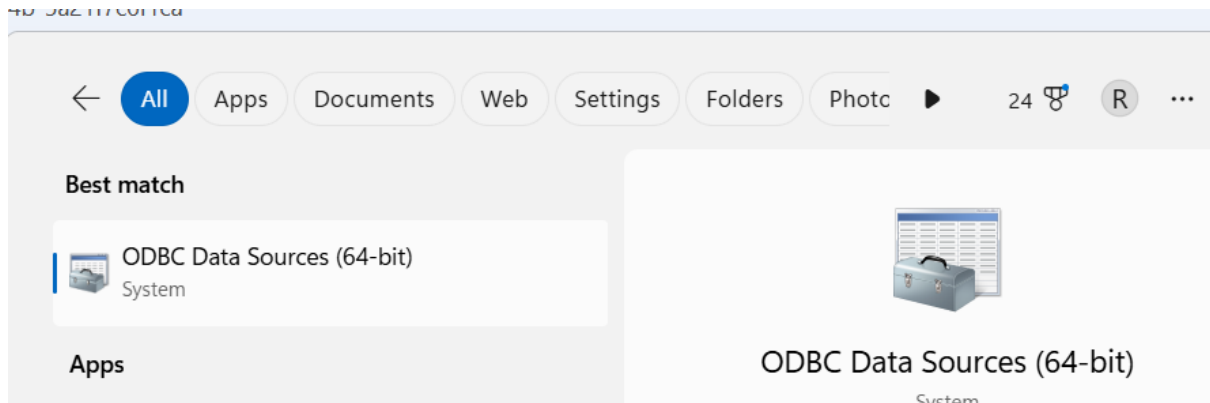
✓ **Practical Example from Your Project**

In your setup:

Power BI → ODBC → Athena Driver → AWS Athena → S3 Data

Power BI did not directly connect to S3; Athena handled querying, and ODBC handled communication.

Download 64-bit version of ODBC:



Added policies for the member so that they can connect to a common workspace

